

Revision notes: *Failure of Converse Theorems of Gauss Sums Modulo ℓ*

Prepared ahead of sponsorship. Section and equation numbers refer to the current draft. These are read-level findings; I have not rerun the SageMath code, so please verify the computational points before acting on them.

Items are grouped by how much is at stake. The first four are worth resolving before submission.

Will's notes: I'm not sure if this made it in anywhere, but at the beginning of your abstract I would state the assumption "Let F_q be a finite field of characteristic p , and let ℓ be a prime number such that $\gcd(\ell, q) = 1$ " or possibly "...such that ℓ divides $q-1$ " which is stronger, and I think necessary for your assumptions anyways?

A. Make the scope of the conjecture explicit (resolved, but state it)

Where: Section 5, "Conjecture 1"; abstract; Section 1; Section 2.5.

The source paper opens with "Let F_q be a finite field of order q and characteristic p , and let ℓ be a prime different from p ." That is the standing hypothesis for the whole paper, so q ranges over all prime powers and the conjecture inherits that scope. Both $(\ell, q) = (2, 49)$, with $p = 7$, and $(3, 4)$, with $p = 2$, satisfy $\ell \neq p$ and are covered. **The disproof stands.**

Two things still to do:

- Say this explicitly in Section 5, immediately before presenting the $q = 49$ counterexample. Every previously published counterexample has q prime, so a reader will silently wonder whether proper prime powers were ever intended. One sentence quoting the source's standing hypothesis removes the doubt before it forms.
- Cite the published version. The work has appeared in *Documenta Mathematica* (Bakeberg, Gerbelli-Gauthier, Goodson, Iyengar, Moss, Zhang), and Conjecture 1 should be pinned to its numbered statement there rather than to the 2023 preprint. Confirm the conjecture is numbered and worded the same way in the published text, since that is the version a referee will open.

The same line also fixes a smaller problem. Section 2.5 states the hypothesis as $\gcd(\ell, q) = 1$. The source states it as ℓ prime and different from p , which is cleaner and is the phrasing to adopt. See also item H on enforcing it in the code.

B. `x.trace()` is not the trace map described in the paper

Where: Section 3, the `trace` method and the paragraph following it; equation (7); Definition 2.4.

Equation (7) states $\text{Tr}_{\{F_{q^2}/F_q\}}(x) = x + x^q$, and Definition 2.4 defines Tr as the map $F_{q^2} \rightarrow F_q$. But `x.trace()` in Sage returns the **absolute trace**, that is, the trace down to the prime field F_p . These coincide exactly when $q = p$ is prime. They differ whenever q is a proper prime power, which is precisely the case at $q = 49$ ($p = 7$) and $q = 4$ ($p = 2$), the two new counterexamples.

The computation is still valid, but the paper does not say why. By transitivity of the trace,

$$\psi_p \circ \text{Tr}_{\{F_{q^2}/F_p\}} = (\psi_p \circ \text{Tr}_{\{F_q/F_p\}}) \circ \text{Tr}_{\{F_{q^2}/F_q\}},$$

and $\psi' := \psi_p \circ \text{Tr}_{\{F_q/F_p\}}$ is a nontrivial additive character of F_q , since $\text{Tr}_{\{F_q/F_p\}}$ is surjective and ψ_p is nontrivial. Theorem 2.5 fixes an arbitrary nontrivial additive character, so the sum the code computes is a legitimate instance of the theorem's setup, just with a specific ψ' rather than the one the reader is picturing.

Recommendation: keep the code as it is and fix the prose. Sage does offer a relative signature, `x.trace(K)`, but switching to it is a bad trade here:

- It returns an element of F_q , and the code uses the trace as an exponent on a generator of order p . An element of F_q cannot index that when q is a proper prime power, which is the only case where any of this matters. You would have to compose with a further map $F_q \rightarrow F_p$, and the natural choice rebuilds the absolute trace by hand.
- Because transitivity is an identity of maps rather than an approximation, the rewritten code computes exactly the same values. No table changes.
- The shortcut version is actively dangerous. If you compute the relative trace directly as `x + x**q`, the result lies in F_q but is still typed as an element of F_{q^2} , so calling `.trace()` on it returns $\text{Tr}_{\{F_q/F_p\}}(2y) = 2 \cdot \text{Tr}_{\{F_q/F_p\}}(y)$. In characteristic 2 that vanishes identically, ψ becomes trivial, and every Gauss sum collapses. Characteristic 2 is where $q = 4$ lives, and nothing in the program would flag it. Coercing into $\text{GF}(q)$ first avoids this, but it is an easy step to miss.

So: rename the method to `absolute_trace`, add a one-line docstring stating that it returns $\text{Tr}_{\{F_{q^2}/F_p\}}$, and put the transitivity argument in the paper beside equation (7). Adjust the surrounding prose so it no longer claims the code evaluates $x + x^q$. The misleading part is the name, not the arithmetic. (If you do use `.trace(K)` anywhere, confirm the signature on both GF(2401) and GF(16) first, since availability depends on the element implementation class and those two use different backends.)

Two additions while you are in this part of the paper:

- `additive_character_generator ** self.trace(x)` raises a field element to a power indexed by an element of F_p , relying on the implicit lift to $\{0, \dots, p-1\}$. One sentence noting that the generator has order p , so only the residue matters, closes the gap.
- Replacing ψ by ψ_c multiplies the θ row of the table by $\theta(c)^{-1}\alpha(c^2)^{-1}$. Any two characters satisfying the restriction hypothesis agree on $c \in F_{q^2}^\times$, so the factor is common to both rows and the collisions are unchanged. One sentence to this effect shows the results do not depend on which nontrivial additive character was chosen, which forestalls an obvious referee question.

C. The Carmichael function is misidentified, and it is probably why the large cases fail

Where: Section 3.1.2, the sentence beginning "The smallest such c is called the multiplicative order of ℓ modulo N' "; equation (9); the `fL_bar_gauss_sum_table` function; Section 5.2 and Table 12.

The text says the smallest c with $\ell^c \equiv 1 \pmod{N'}$ "is exactly what the Carmichael lambda function, denoted $\lambda(N')$, computes." This is false. $\lambda(N')$ is the exponent of the group $(\mathbb{Z}/N')^\times$, that is, the smallest positive integer that works simultaneously for every unit. The multiplicative order of ℓ modulo N' divides $\lambda(N')$ and is usually far smaller.

Setting $c = \lambda(N')$ is still **sufficient**, since $\text{ord}(\ell) \mid \lambda(N')$ implies $\ell^{\lambda(N')} \equiv 1$, so $F_{\{\ell^c\}}$ does contain all N' -th roots of unity. Every value in every table is correct. Only the minimality claim is false. There are two ways forward, and they cost very different amounts.

Option 1: fix the sentence, change nothing else

Rewrite the claim so that $\lambda(N')$ is presented as a valid choice rather than the minimal one: the multiplicative order of ℓ modulo N' divides $\lambda(N')$, so $\ell^{\lambda(N')} \equiv 1$ and the required roots of unity are present. Tables 1 through 11 stand, the code is untouched, and Table 12 keeps its -1 rows.

Section 5.2 then needs an accurate account of why those rows failed, which is the oversized field described below, not factorization failures.

This is a two-sentence edit and is enough to make the paper correct.

Option 2: use the actual order and rerun

Replace `carmichael_lambda(N_prime)` with `Mod(1, N_prime).multiplicative_order()` in `fL_bar_gauss_sum_table`. There is no efficiency argument against this; Sage computes the order by factoring the group order, and N' has already been factored in spirit by stripping the ℓ -part.

The payoff is Section 5.2. Consider the failed case $(\ell, q) = (2, 257)$:

quantity	value
$q^2 - 1$	$66048 = 2^9 \cdot 3 \cdot 43$
$N = p(q^2 - 1)$	$257 \cdot 66048$
$N' = N / 2^9$	$33153 = 3 \cdot 43 \cdot 257$
$\lambda(N') = \text{lcm}(2, 42, 256)$	5376
$\text{ord}_{\{N'\}}(2) = \text{lcm}(2, 14, 16)$	112

The gap comes from the 43 and 257 factors: λ takes 42 and 256 whole, while 2 has order only 14 and 16 there, since $2^7 \equiv -1 \pmod{43}$ and $2^8 \equiv -1 \pmod{257}$. The current code therefore asks Sage for $\text{GF}(2^{5376})$, where no Conway polynomial exists, so it falls back to a random irreducible polynomial and every multiplication becomes a degree-5376 reduction. $\text{GF}(2^{112})$ is a different proposition entirely. With $\text{theta_range} = 66048/2^9 = 129$ and $\text{alpha_range} = 256/2^8 = 1$, the table is roughly 129×66048 field operations.

Please verify these numbers before relying on them. Several of the -1 rows in Table 12 may then fill in, which would turn the weakest section of the paper into a result.

If you do rerun, expect relabeled indices. Changing c changes the field, hence the generator h , hence which primitive N' -th root of unity ζ is used. Replacing ζ by ζ^k relabels θ_i as $\theta_{\{ik\}}$, so a collision class currently printed as $\{1, 7, 43, 49\}$ may come back as a different set of four indices. The partition structure, every reported group size, and the counterexample status are all unchanged. The discrete logarithms themselves are stable, since `finite_field.gen()` depends only on q . Do not read a relabeled table as a contradiction of the old one.

Related: h is not guaranteed to be a multiplicative generator

Section 3.1.3 says h is "a generator in the multiplicative group of F_{ℓ^c} ," but the code sets $h = F.gen()$, which returns a root of the defining polynomial. That is a multiplicative generator when Sage builds the field from a Conway polynomial, and not otherwise. Past the Conway database Sage uses a random irreducible polynomial, whose root need not be primitive. If it is not, $h^{**}(1^{**}c - 1) // p$ need not have order p and the additive character silently degenerates.

This only bites at large c , which is precisely where the -1 rows are, so it may be a second cause of trouble there alongside the field size. Either assert $h.multiplicative_order() == 1^{**}c - 1$ or use $F.multiplicative_generator()$. Worth doing under either option above.

Either way

State in Section 3.1.2 that the argument requires $\gcd(\ell, N') = 1$, so that ℓ is a unit modulo N' and has an order at all. This holds because Step 1 removed every factor of ℓ from N , but the reader currently has to supply it.

D. The restriction criterion is correct, but needs to be proved

Where: Section 3, the paragraph describing `partition_by_mod_q_minus_1_power`; Section 4, the displayed congruence.

The claim is that $\theta_i|F_{q^a}^\times = \theta_j|F_{q^a}^\times$ holds exactly when the θ -indices are congruent modulo $(q-1)/\ell^a m$. This is the step that separates genuine counterexamples from the trivial collisions the classical theorem already predicts, so it carries the whole argument, and at present it is asserted rather than derived.

I checked it and **it is true**. It just needs to appear in the paper as a lemma with proof. Here is the argument.

Let $a = v_\ell(q^2 - 1)$ and $M = (q^2 - 1)/\ell^a$, so M is `theta_range`. Let $m' = v_\ell(q - 1)$ and $Q = (q - 1)/\ell^{m'}$, so Q is `mod_base`. Fix a generator g of $F_{q^2}^\times$, note that g^{q+1} generates $F_{q^a}^\times$, and let ζ be the multiplicative character generator, of order exactly M , so that $\theta_i(g^t) = \zeta^{it}$.

Lemma. For $0 \leq i, j < M$, we have $\theta_i|F_{q^a}^\times = \theta_j|F_{q^a}^\times$ if and only if $i \equiv j \pmod{Q}$.

Will's Note: the proof given above is completely fine afaik and you're welcome to use it, but it's also something you could prove yourself, if you'd rather write it in your own words. Clear formatting to reveal.

Note what the proof consumes: ζ must have order **exactly** M . Section 3.1.3 currently establishes only that its order divides N' , which is not enough. The exact order is $N'/p = M$, and establishing that requires h to be a primitive element. This is the same gap flagged at the end of item C, so the assertion suggested there is not merely defensive coding; this lemma depends on it. Add the order computation to Section 3.1.3.

Two related fixes:

- Section 4 writes the modulus as $(q - 1) // \ell$, while the code uses $(q - 1)/\ell^{m'}$. These disagree whenever $m' > 1$, which includes $q = 49$, where $m' = 4$ and $Q = 3$. Correct the text to match the code.
- Section 3, in the discussion of `find_counterexamples`, states that any grouping of three or more θ -indices "must constitute a true counterexample." As written this is stronger than the classical theorem supports, since the conclusion is conditional on the restriction hypothesis. The sharp statement is available now: in index terms $\theta_i^q = \theta_{iq \bmod M}$, so the collisions the classical theorem predicts are exactly the pairs $\{i, iq \bmod M\}$. A counterexample is a class containing three or more indices that all lie in one residue class mod Q .

This also gives a much better way to present the headline result. At $(\ell, q) = (2, 49)$ we have $M = 75$ and $Q = 3$, and the class $\{1, 7, 43, 49\}$ consists of the two Frobenius pairs $\{1, 49\}$ and $\{7, 43\}$, since $1 \cdot 49 = 49$ and $7 \cdot 49 \equiv 43 \pmod{75}$, fused into a single class that lies entirely within one residue class mod 3. Saying it that way is far more convincing than observing that the group size exceeds two.

E. The third column of every results table is generated incorrectly

Will's Note: if you include the proof of D, you could simplify this column to a single value, the common residue modulo $(q-1) / \ell^m$ of all elements in the subgroup. It would be easy to verify that the elements listed indeed match the stated residue, and column four could be replaced by

a single True/False that verifies this. It could even be one combined column, just make sure the code is evaluating that every element in the subgroup is actually the same residue, then make that clear at a glance to the reader.

Where: the `__main__` block; Tables 1 through 11.

The line

```
mod_subgroup_str = list_to_curly_str(sorted(sum(mod_subgroups, [])))
```

flattens the partition back into a single list before printing it. So the column headed $\theta_1|F^*_q = \theta_2|F^*_q$ is just column 1 re-sorted, in every table. That is why the two columns are identical everywhere, which a reader will notice and wonder about.

`subgroups_to_curly_str` is defined in Section 3 and given a full paragraph of explanation, including the worked $q = 97$ example, but is never called anywhere in the program.

Fix: either call `subgroups_to_curly_str(mod_subgroups)` so the column shows the actual partition, or drop the column and keep only the size in column 4. Column 4 uses `max(len(s) for s in mod_subgroups)`, which is computed from the real partition, so **the results themselves are unaffected**. This is a display bug, not a mathematical one, but it should be fixed before a referee sees it.

F. Statement and notation

- **Definition 2.2.** The multiplicative character is introduced as α , then the next sentence says "We extend χ to all of F_q by setting $\chi(0) := 0$," and the following sentence also uses χ . Pick one symbol.
- **Definition 2.2.** Say the additive character is a homomorphism *from the additive group* $(F_q, +)$ to $K^\times$, since as written the two bullets look like they have the same source and target.
- **Definition 2.4.** Uses Θ , α , Ψ in the statement but ψ and `tr(x)` in equation (2). Unify capitalization, and use Tr throughout to match equation (7). Equation (2) also ends in a comma with no following clause.
- **Theorem 2.5.** The hypothesis lets α range over characters of $F_{q^m}^\times$ and equation (4) writes $\alpha(a)$ for $a \in F_{q^m}^\times$, but everywhere else, including Definition 2.4 and the actual computation, α is a character of $F_q^\times$ evaluated at $N(x)$. State the theorem in the same notation you use in Definition 2.4, or add a sentence reconciling the two.
- **Section 2.5.** Says the identity holds "for all additive characters ψ of F_{q^2} ," but Theorem 2.5 fixes a single nontrivial ψ . Make these agree.
- **Section 3.1.3.** Stray "2" in "polynomial $x^3 - 1$ ". 2These two values."

- **Sections 3, 4.** Bare equation references read as numbers: "a twisted Gauss sum of the form 2," "i.e., 3," "should imply 5." Write these as (2), (3), (5).
 - **Table 1.** The grouping $\{0, 1, 2\}$ for $(\ell, q) = (2, 5)$ includes $\theta = 0$, the trivial character. A sentence on whether the classical conclusion is meaningful when one member of a collision class is trivial would preempt an obvious question.
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G. References

- **[1]** is published: Bakeberg, Gerbelli-Gauthier, Goodson, Iyengar, Moss, Zhang, *Documenta Mathematica*. Cite the published version rather than "preprint, 2023," and pin the conjecture to a numbered statement there.
 - **[8]** is published as Nien and Zhang, *Converse theorem of Gauss sums*, J. Number Theory **221** (2021), 365–388, with an appendix by Zhiwei Yun. The draft cites the 2018 arXiv preprint under its earlier title. Theorem 2.5 is attributed to "[8, Theorem 1.2]," so the numbering should be checked against the published paper.
 - **[9]** Pinner, *Twisted Monomial Gauss Sums Modulo Prime Powers*, is cited as the source for the notion of a twisted Gauss sum. Pinner's twisting is a different construction, involving monomial arguments modulo prime powers, not the F_{q^2}/F_q twist by $\alpha \circ N$ used here. This looks like a mis-citation. Either drop it or replace it with the source you actually took the definition from, most likely [8]. It also carries no date.
 - **[3]** and **[4]** are course notes and a lecture PDF. For standard facts about finite fields and characters, cite a textbook such as Lidl and Niederreiter, *Finite Fields*, or Ireland and Rosen.
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H. Presentation of the computational contribution

- **No code availability statement.** The program *is* the contribution, and right now it exists only as code fragments interleaved with prose. Add a repository or archive link and record the SageMath version used. Referees increasingly ask for this, and it is the cheapest way to make the results checkable.
- **Table 11** is described as partial "due to formatting constraints." A partial table cannot be verified. Move the full output to an appendix or the repository and say where it is.
- **Table 12** says "first 17 entries" without saying how the (ℓ, q) pairs were enumerated or where the list was cut off. State the search range.
- **Abstract and Section 1** mention only (2,49). If A resolves in your favour, mention (3,4) and (2,97) as well, since three counterexamples of differing shape make a stronger case than one.
- **fL_bar_gauss_sum_table** validates that ℓ is prime and q is a prime power, but never checks $\gcd(\ell, q) = 1$, which Section 2.5 assumes. Add the check and raise on failure.

Suggested order of work

1. Items B and E. Both are small edits and both are things a referee will read as errors.
2. Item C, at least Option 1. Option 2 and a rerun of Table 12 only if you have time before the deadline.
3. Item D, as a lemma.
4. Item A, which is now a sentence in Section 5 and a citation update.
5. Items F through H as a cleanup pass.